

Advanced Deep Learning Technique for Digital Predistortion to Improve Power Amplifier Linearity

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Abstract—Power amplifiers (PAs) are crucial in RF RADAR and communication systems, but their nonlinearities often degrade signal quality. This paper presents advanced deep learning techniques, specifically 1D-Convolutional Neural Networks (1D-CNNs) and Long Short-Term Memory (LSTM) networks, to enhance digital pre-distortion (DPD) and improve PA linearity. The method addresses complex nonlinear distortions more effectively than traditional models based on memory polynomials by using CNNs for spatial feature extraction and LSTMs for temporal dependencies. Experimental results show significant improvements in signal fidelity, with reduced inter-modulation distortion (IMD) and enhanced adjacent channel power ratio (ACPR). This study highlights the potential of deep learning to advance DPD techniques by comparing results obtained from conventional DPD methods and optimizing PA performance in modern RADAR and communication systems.

Index Terms—Digital Predistortion (DPD), Deep Learning, Convolutional Neural Networks (CNN), Long Short-Term Memory Networks (LSTM), Power Amplifier (PA), Nonlinearity Compensation

I. INTRODUCTION

The growing need for high-speed data transfer in RADAR and wireless communication demands advanced modulation techniques like Non-linear Frequency Modulation (NLFM), Quadrature Phase Shift Keying (QPSK), and Orthogonal Frequency Division Multiple Access (OFDMA). These technologies require high-performance RF components, such as power amplifiers to get a higher coverage range. However, high-power amplifiers' nonlinearities can result in IMD and adjacent channel interference, affecting system efficiency [1].

Traditional methods [2], such as memory polynomial structures, offer limited improvements in compensating for these nonlinearities. Conventional DPD techniques attempt to counteract these effects by applying inverse characteristics to the input signal. Recently, the application of deep learning [3] has shown promise in power amplifier modeling, offering new approaches to achieve linearization. Deep learning models, intense neural networks (DNNs), recurrent neural networks (RNNs), and CNNs, have been developed to enhance power amplifier performance. The proposed DPD framework is shown in Fig.1. It employs a digital environment where a modulated signal is pre-distorted using a deep learning model to counteract PA non-linearities. The signal passes through an RF DAC, is amplified by the PA, and then assessed via a feedback loop

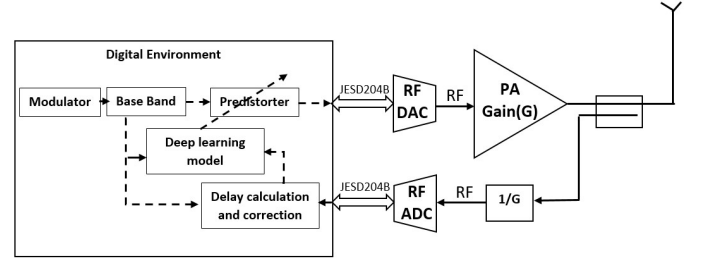


Fig. 1. DPD framework based on deep learning models

using an RF ADC. The feedback, estimation and correction of delay, are given to the deep learning model to adjust pre-distortion in real time, optimizing transmission quality.

II. CONCEPTUAL DESIGN

A. Time delay calculation of PA due to memory effect

[4]

Due to the memory effects and structural design of power amplifiers, there is an inherent delay in their response. A time delay estimation technique, utilizing cross-correlation, is employed to achieve precise predistortion. This approach evaluates and corrects the delay between the power amplifier's input (x) and output signals (y).

The cross-correlation of two discrete-time signals $x[n]$ (PA input signal) and $y[n]$ (PA output signal) is given by:

$$R_{xy}[k] = \sum_{n=-\infty}^{\infty} x[n] \cdot y[n+k] \quad (1)$$

where: $R_{xy}[k]$ is the cross-correlation sequence, $x[n]$ is the input signal, $y[n]$ is the output signal, and k is the lag or shift applied to the output signal.

The lag $k_{\max}$ that maximizes the cross-correlation $R_{xy}[k]$ indicates the best alignment between the input and output signals, which corresponds to the time delay.

$$k_{\max} = \arg \max_k R_{xy}[k] \quad (2)$$

If the signals are sampled at a sampling frequency f_s , the time delay Δt is given by: $\Delta t = \frac{k_{\max}}{f_s}$. In other ways, equation 2 can be written as

$$Z(K_1) = \max[(x * y)(\theta)] \quad (3)$$

So K_1 is the lag or shift parameter corresponding to $x(n)$ and $y(n)$ cross-correlation magnitude is maximum.

Corrections to the delay for the DPD algorithm

$$Xcorr[n] = f(1 : end) \quad (4)$$

$$Ycorr[n] = y(1 + K_1 : end) \quad (5)$$

PA out and input is captured and delay is calculated in Mathematical as shown in Fig. 2. The first plot of the Fig 2 shows the delay of input and output and the second plot shows the after-the-delay compensation.

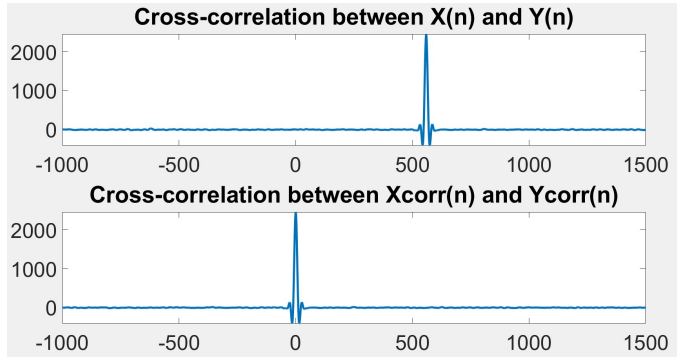


Fig. 2. Cross correlation of Input and PA output

B. Memory Polynomial Model for DPD

The memory polynomial model is a widely used traditional approach to model the PA's nonlinearity, expressed as [6]:

$$y(n) = \sum_{m=0}^M \sum_{k=0}^K a_{km} \cdot x(n-m) \cdot |x(n-m)|^k \quad (6)$$

where a_{km} are the polynomial coefficients, M is the memory depth, and K is the nonlinearity and n is the time index.

DPD is the process of find the inverse polynomial $f^{-1}(x_1, x_2, x_3, \dots, x_n)$ function of the PA model is

$$Xcorr_{mk}(n) = \sum_{k=0}^{K-1} \sum_{m=0}^{M-1} b_{km} \cdot Ycorr(n-m) \cdot |Ycorr(n-m)|^k \quad (7)$$

where $Xcorr$ and $Ycorr$ are calculated using equations 5 and 6, b_{km} are the inverse polynomial coefficients which are calculated using least mean square(LMS) algorithm [5] in adaptive DPD.

III. PROPOSED DEEP LEARNING APPROACH

Deep learning models are now widely used for learning the parameters and hence doing the corrective measure. This is research, there are two models have been proposed as follows

A. Convolutional Neural Networks and Long Short-Term Memory Networks (CNN-LSTM) model

CNNs are effective at extracting spatial features from input signals. CNNs can model the frequency-domain characteristics of signals passing through the PA. LSTMs capture temporal dependencies and are well-suited for modeling the dynamic behavior of PAs over time. The proposed CNN-LSTM model is shown in Fig.3.

The model is divided into an input layer, a 1D-convolutional layer, an LSTM layer, and a fully connected layer.

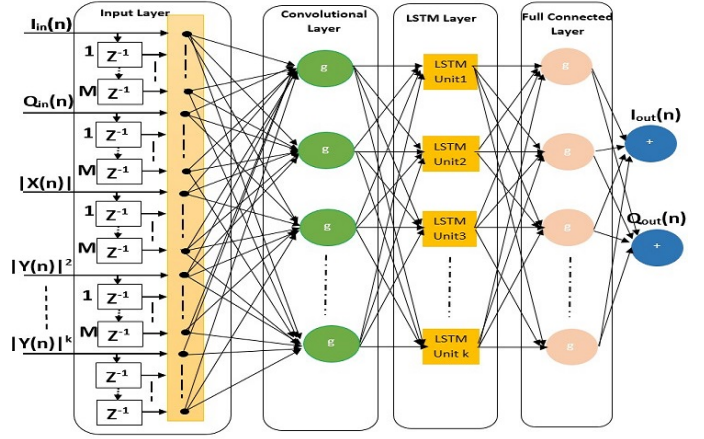


Fig. 3. CNN-LSTM based deep learning model

1) *Input Layer*: The input layer inputs the (I_{in}/Q_{in}) of the baseband signals. The I_{in}/Q_{in} samples are delayed depending on the memory effect of the in PA model and the magnitude of the I_{in}/Q_{in} samples up to the degree K^{th} of the power are fed as input to the model.

2) *1D-Convolutional layer*: The backpropagation algorithm is a fundamental method for training CNNs. It involves computing the gradient of the loss function for each weight in the network and updating the weights to minimize the loss. For a convolutional layer, the operation is:

$$z_{i,j}^{(l)} = (W^{(l)} * a^{(l-1)})_{i,j} + b^{(l)} \quad (8)$$

where: $W^{(l)}$ is the filter (kernel) matrix for layer l , $a^{(l-1)}$ is the activation map from the previous layer, $b^{(l)}$ is the bias term for layer l , $*$ denotes the convolution operation, $z_{i,j}^{(l)}$ is the pre-activation value at position (i, j) in layer l .

The activation function (ReLU) is applied to $z^{(l)}$ to get the activation $a^{(l)}$:

$$a_{i,j}^{(l)} = \sigma(z_{i,j}^{(l)}) \quad (9)$$

3) *LSTM layer*: LSTM networks are a type of RNN designed to capture long-term dependencies. Forget Gate, Input Gate, Cell State Update, and Output Gate equations are as below $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$, $i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$, $\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$, $C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$, $o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$, $h_t = o_t \odot \tanh(C_t)$

Where: x_t is the input at time step t from CNN layer, o_t is the output gate

4) *Fully connected layer*: The forward pass on each neuron of the Fully Connected Layer is as simple as this,
 $\text{output} = \Phi(W_{ij} \cdot X_j + b_j)$ Where: W = Weights, X = inputs from the preceding layer, Φ = Activation function.

B. CNN based deep learning model

One of the simplest models of deep learning is CNN as shown in Fig. 4.

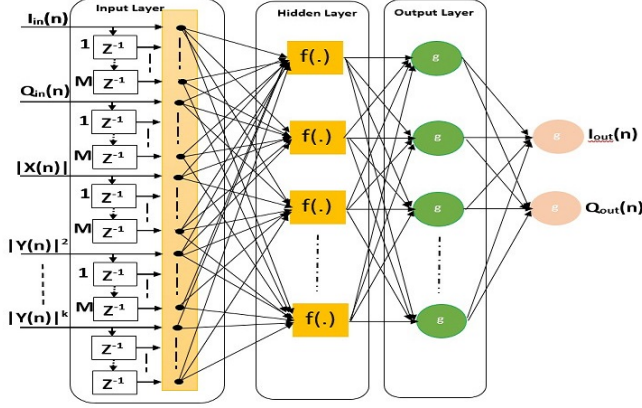


Fig. 4. CNN-based deep learning model

Both the deep learning models are simulated with real-time PA data and found that CNN-LSTM is better than CNN. So further all the results are limited to the CNN-LSTM model only.

IV. IMPLEMENTATION

The DPD test setup, shown in Fig. 5, was designed for an L-Band signal with a bandwidth of 10 MHz. Mathematical was utilized to generate the baseband signal, which was then transmitted as IQ data to an FPGA over a 1G Ethernet connection. A direct RF sampling transceiver was used to convert the baseband signals to RF before passing them to the power amplifier (PA). After being attenuated, the PA output was fed back to the digital transceiver for conversion into a digital stream. This stream was subsequently sent back to Mathematical via the 1G Ethernet. The power amplifier's transfer characteristics were plotted, as shown in Fig. 6.

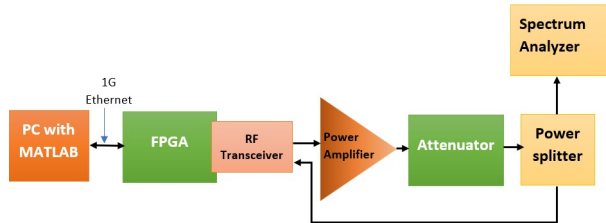


Fig. 5. LAB Test setup for DPD experiment

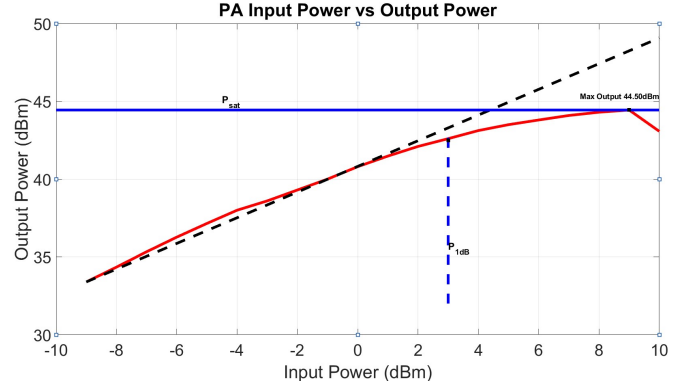


Fig. 6. PA transfer characteristic

TABLE I
PA PARAMETERS USE IN THIS RESEARCH

Parameter	Value
Power amplifier linear gain	41.9 dB
1 dB compression input power (P1dB)	-1.5 dBm
PA maximum output power (Psat)	44.47 dBm
Maximum correctable input power	2 dBm

Using the time delay calculation mentioned in section II-A of input and output, the transfer function of PA and its inverse transfer function is analyzed and shown in Fig.7. In this experiment, the power amplifier's degree of nonlinearity and memory effect are considered to be $M=5$ and $K=6$, respectively.

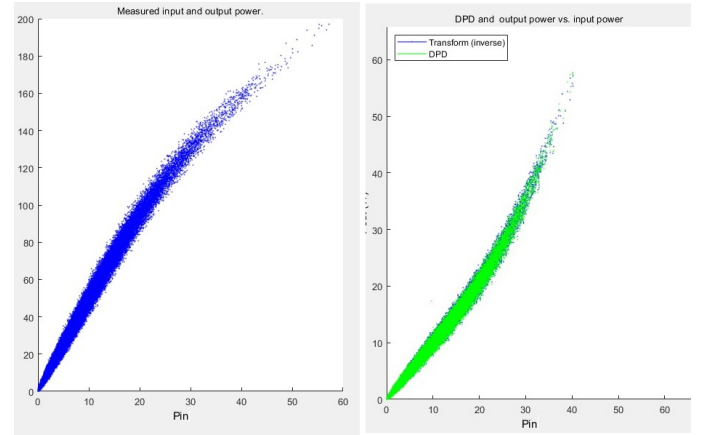


Fig. 7. PA transfer characteristic and pre-distorter transfer characteristic

V. RESULTS

As explained in section IV above the test setup, experiments are conducted for the memory polynomial-based solution and also with deep learning models.

A. Memory polynomial based DPD

Considering the $M=5$ and $K=6$, memory polynomial-based DPD has been implemented. Memory-based DPD output is shown in Fig. 8. IMD improvement was found to be 13.0 dBc.

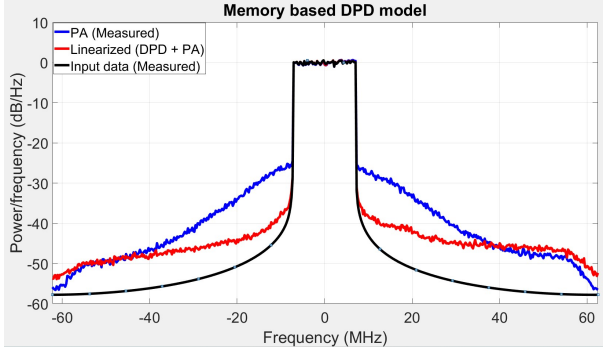


Fig. 8. Mathematical simulation for Memory polynomial based DPD

B. Deep learning polynomial based DPD

CNN-LSTM model is trained and DPD performance has been analyzed. Training data is collected from PA input-output pairs, with the CNN-LSTM model trained to predict inverse distortion characteristics. The hybrid CNN-LSTM model integrates spatial and temporal learning for improved DPD performance and the result is shown in Fig. 9. IMD improvement is found to be 21.2 dBc.

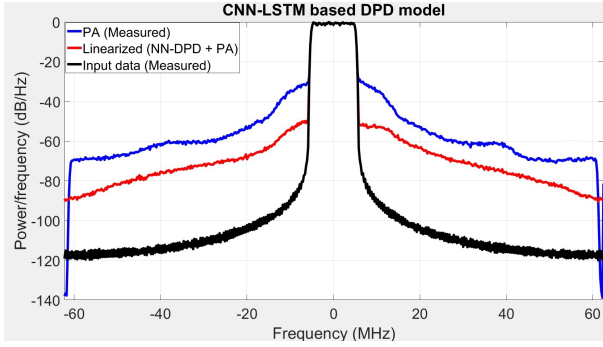


Fig. 9. Mathematical simulation for CNN-LSTM based DPD

DPD model has been tested in real-time and the PA's output spectrum without DPD is shown in Fig. 10 and with DPD is shown in Fig. 11.

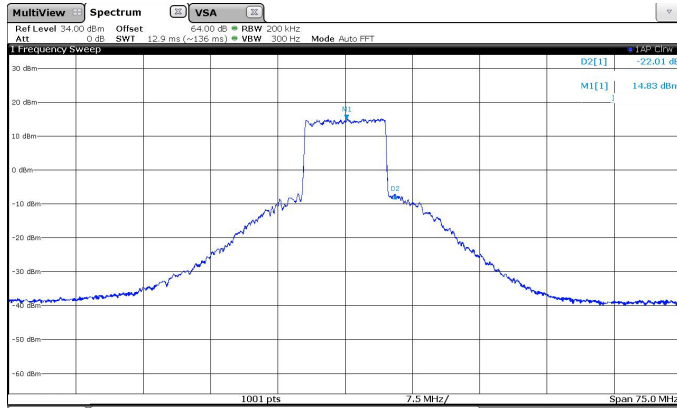


Fig. 10. Real-time PA's output spectrum without DPD

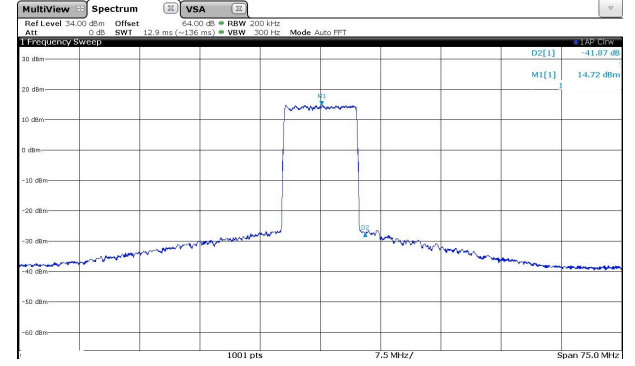


Fig. 11. Real-time PA's output spectrum with DPD

The final comparison of the models is shown in Table II.

TABLE II
COMPARISON OF IMD VALUES

Training models	Mathematical simulation (IMD (dBc))	Real time (IMD (dBc))
Without DPD	-27	-22.01
DPD with memory model	-40	-35.41
DPD using CNN-LSTM model	-49.3	-41.87

VI. CONCLUSION

Simulations and experiments show that the CNN-LSTM-based DPD system greatly improves IMD levels compared to traditional memory polynomial methods. The adaptability of deep learning helps the system perform well in different signal environments, ensuring consistent improvements.

This paper covers how time delays are estimated and corrected to address the memory effects of power amplifiers (PAs). It briefly discusses the traditional memory polynomial-based DPD model and its implementation and a deep learning-based approach for digital pre-distortion in power amplifiers using the CNN-LSTM model to address nonlinear distortion. The proposed system demonstrates that the CNN-LSTM model performs better than conventional methods, representing a significant step towards more efficient RADAR and communication systems.

ACKNOWLEDGMENT

I express my gratitude to Mr. Venkatamuni T (DGM, RFMW-PDIC) and Devindra M C (DGM, Automation Solutions-PDIC) for sharing their expertise and for providing the power amplifier during the development of this new technology project. Additionally, I would like to extend my thanks to Mrs. Seema (DGM, CoE-Radar and Weapon Systems) and Mrs. Dharani B, Chief Technology Officer, CoE-Radar and Weapon Systems, for her continuous support and for providing me with the opportunity to work on this new technology project.

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